Round 13
Carry pair innovations

One mechanism: a fourth-cumulant birth model transported with the existing cubic-source propagators.

Status: a concrete, cheap hypothesis, **not a validated correlation ≥0.7 solution [Q: requested gate]**. The transport and incremental FLOP ledger are exact for the stated inputs. Accuracy of the birth approximation is unknown.

Number labels: Q = quoted from your prompt; D = derived here; E = estimated. All numerical coefficients and operation counts in equations are D; n=1024, L=16 and 1 u=2^{31} FLOPs are Q. No network result is claimed as measured.

Object: keep the birth metric's transported form

D. Attach two signed vectors γ_b, η_b ∈ R^n to every existing cubic source. Reuse its propagator P_b ∈ $R^{n×n}$; do not create another dense leg. Define two implicit matrices and the carried tensor:

$$H_b = P_b \mathrm{diag}(\gamma_b) P_b^T, \qquad S_b = P_b P_b^T,$$

$$\widehat{K}_b = \mathcal{S}_6(H_b, S_b) + \sum_r \eta_{br} p_{br}^{\otimes 4}.$$

D. Here p_br is a column of P_b and S_6 is the unnormalized six-term symmetric product:

$$[\mathcal{S}_6(H,S)]_{ijkl} = H_{ij}S_{kl} + H_{ik}S_{jl} + H_{il}S_{jk} + H_{jk}S_{il} + H_{jl}S_{ik} + H_{kl}S_{ij}.$$

D. At birth P_b=I. Thus the birth's off-diagonal K22 entries equal γ_bi+γ_bj, its marginal fourth cumulant equals 6γ_bi+η_bi, and its other index sectors vanish. After mixing, H_b and S_b are dense: the tensor has a genuine all-distinct sector. Neither matrix is formed explicitly.

D: transport. With your row-vector convention, B_l=W_l^{T}diag(Φ_l). For sources already present before the ReLU,

$$P_b^+ = B_l P_b, \qquad \gamma_b^+ = \gamma_b, \quad \eta_b^+ = \eta_b, \qquad \widehat{K}_b^+ = B_l^{\otimes 4} \widehat{K}_b.$$

D. The newborn is inserted after the ReLU with P_b=I and crosses the raw W_l^T. These are the propagators already computed for K_3. Under your existing old-leg projections, the tensor remains exactly the polynomial of the retained P; the additional κ_4 error caused by those projections is untested.

Why the closed-list measurements do not directly test this

D. This carries the complete modeled tensor from every birth, including its off-diagonal sectors; it is not a diagonal-state update using W^4. There is no new leg-rank truncation or source cap. It is also not the covariance-only predictor a·Sym(C_l⊗C_l): each S_b retains its own weight history, H_b has independent signed weights, and η_b is a separate signed tensor term. Its quartic form can be negative, unlike a centered Gaussian covariance-mixture's fourth cumulant.

D. The approximation is instead explicit and local: project the *nonlinear birth's* K22 onto a row-additive form, match its marginal birth exactly, and discard the other birth sectors. A test of this same full-history, shared-leg birth model would close this proposal; low-rank leg tests or a current-C scale-mixture test do not establish its error.

Birth rule and diagonal readout

D. Use the chain's own modeled pre/post K22 and marginal fourth cumulants, not teacher values. Denote these by bars and put d_i=Φ_i, u_i=d_i². Subtract the modeled linearly carried part to define the nonlinear innovation:

$$J_{ij} = \bar{K}22_{ij}^{post} - u_i u_j \bar{K}22_{ij}^{pre} \quad (i \neq j),$$

$$b_i = \bar{k}_{4,i}^{post} - u_i^2 \bar{k}_{4,i}^{pre}.$$

D. Let J_ii=0. The unique unweighted least-squares fit of J_ij by γ_i+γ_j over off-diagonal pairs is

$$r_i = \sum_{j \neq i} J_{ij}, \qquad s = \frac{\sum_i r_i}{2(n-1)}, \qquad \gamma_i = \frac{r_i - s}{n-2}, \qquad \eta_i = b_i - 6\gamma_i.$$

D. Hence the modeled birth matches b_i exactly. This is an analytic moment projection, with no trained coefficient or λ refit. Subtracting the linear part avoids repeatedly injecting the whole regenerated κ₄ level. These innovations use the chain's non-Gaussian inputs; they are not a Gaussian-input estimate of the entire deep κ₄. Keep the chain's regenerated core and its pair closure as the innovation provider; the new readout replaces its κ₄ only in the mean closure.

The existing core makes the birth calculation linear in n

D, conditional on the stated core. Write v_i= k̄₄,pre_i and k_i=v_i/6. For Sym(G⊗mI), the off-diagonal pre-K22 is k_i+k_j. If R_i is the already computed post-K22 off-diagonal row sum, then no new n×n birth array is needed:

$$r_i = R_i - u_i\left(k_i \sum_j u_j + \sum_j u_j k_j - 2u_i k_i\right).$$

D. This uses the same post-K22 row sums that the regenerated-core update can expose. The pre/post channels must use consistent cumulant normalization. If your modified implementation no longer has this pre-K22 form, use the full J formula and charge its explicit array operations; do not silently apply the shortcut.

Readout: matrix-vector work only

D. At the next pre-activation, reuse Z_b=P_b∘P_b from the cubic slice routine. Compute

$$h_b = Z_b \gamma_b, \qquad s_b = Z_b \mathbf{1}, \qquad q_b = 6h_b \circ s_b + (Z_b \circ Z_b)\eta_b, \qquad \widehat{k_4} = \sum_b q_b.$$

D. This is exactly the diagonal of Σ_b K̄_b for the retained legs. The source's all-distinct entries contribute implicitly through the transported Gram forms. Holding the previous state fixed, changing W_l to −W_l leaves this quartic readout unchanged, as required; the odd-parity obstruction for cubic sources does not transfer to this readout.

D. Start with zero pre-activation κ₄ at the first layer, since that pre-activation is Gaussian. Use k̄₄ in the existing Edgeworth mean formula *in place of* the regenerated diagonal, not added to it. Retain all births through the output. This does not change the covariance or K₃ carrier.

Exact incremental FLOP ledger

D. Counts below are additional to the existing chain. Persistent extra state: two n-vectors per birth, giving 122,880 bytes for fifteen births in float32. The underlying P matrices and their re-formation are already paid for by K_3. Evaluate the new readout while those buffers are live; reconstructing P a second time is not included or permitted by this ledger.

Operation [D]	Real shape [D]	FLOPs [D]
s = row_sum(Z)	n × n → n	n(n−1)
h = Zγ	(n,n) @ (n,1)	n(2n−1)
t = (Z∘Z)η	n² multiply + (n,n) @ (n,1)	3n²−n
q = 6h∘s + t	Three vector operations	3n
Per source, total	No matrix-matrix product	6n²
Sum k source outputs	k vectors of length n	(k−1)n
Birth, using the row shortcut	Vector operations/reductions	19n−2

D: birth count. Form k=v/6, u=d², t=k∘u; reduce u and t; evaluate r=R−u∘(kΣu+Σt−2t); form b=$k_{4,\text{post}}$−u²∘v; then compute s, γ and η. This sequence costs 19n−2 operations. The dimension-dependent denominators are precomputed constants. Copies, indexing and diagonal extraction carry no arithmetic charge under the specified model.

$$F_\ell = 6k_\ell n^2 + (k_\ell - 1)n + \mathbf{1}_{\ell<L}(19n-2), \quad k_\ell = \ell - 1 \geq 1.$$

D. The first layer has only the birth charge. The final layer has no new birth charge. Across the network there are 120 source-readouts and fifteen births:

$$F_{\text{added}} = 720n^2 + 390n - 30 = 755{,}374{,}050 \text{ FLOPs},$$

$$F_{\text{added}}/2^{31} = 0.3517484525 \text{ u}.$$

D. The largest layer increment is 0.0439519882 u. Additional dense square products: zero. Additional matrix-vector products: two per live source, charged at their real shapes above. Strassen is not invoked for this added work.

D. If P∘P cannot be reused, charging one n² square operation per source-readout raises the total to 0.4103422025 u. If post-K22 row sums are not reused but its matrix already exists, ordinary row reductions add 0.0073170662 u over the network. If neither pre/post K22 is computed by your modified chain, its closure cost must be included separately; this ledger assumes those existing channels.

Q → D. With the quoted baseline of approximately 232 u, the specified added arithmetic gives approximately 232.352 u total. The baseline is approximate; the incremental integer FLOP count is exact for this schedule. This is an accuracy proposal with negligible added arithmetic, not a claim to halve the baseline cost.

D: the symmetric-product identity, transported diagonal and row-fit normal equations were independently checked on small random arrays. These checks verify algebra only. They are not measurements of network correlation, memory usage or runtime.

Why it is worth testing, and what is not established

Q. Your true linear carry retains 83-90% of the deep level and has pattern correlation 0.77-0.80 on the public network; the pair contribution dominates the reported norm. Those findings motivate preserving the full history of a pair-sector birth model. They do not show that its row-additive projection captures the centered pattern.

D. This carrier keeps precisely the memory that a reset to Sym(G⊗mI) discards: the initially diagonal metric is transported into $S_b=P_bP_b^T$. Its off-diagonal entries remain paired with the transported H_b. No expectation over later weights is taken. The signs and directions of the actual network determine every source's contribution.

Unknown accuracy. The model omits the residual K22 birth after row fitting, the K31 birth, and the all-distinct nonlinear birth. It also uses imperfect chain inputs, and its κ_4 response may be more sensitive than κ_3 to existing old-leg projections. Subtracting two approximate pre/post channels can magnify their errors. The quoted dominance of the total pair norm does not control any of these errors.

D, conditional estimate. Suppose the candidate reproduces an ideal carried prediction of correlation ρ_0 and adds centered error independent of both that prediction and the teacher. Then its correlation is $\rho_0/\sqrt{(1+\tau^2)}$, where τ is added-error standard deviation divided by the ideal predictor's standard deviation. Using ρ_0=0.77-0.80 [Q], reaching 0.7 [Q] requires $\tau\leq$0.458-0.553 [D]. Independence is an assumption, not a measured property; no bound on τ has been established.

Answer to the accuracy requirement: I cannot claim that this reaches the requested gate. Its cost and transport are established; whether these particular innovations retain the required pattern is the unresolved hypothesis.

One falsifier: a free-history replay on both networks

Protocol. Run the carrier from the first layer with the chain's own stored pre/post κ_4 channels and gains, and with the actual P buffers used by the cubic chain. Teachers must never enter birth, transport, normalization or coefficients. Retain every birth; use the existing old-source approximations exactly as shipped. This tests a free κ_4 carrier driven by frozen baseline channels, not a teacher-forced step and not yet a fully coupled mean rerun.

Q: gate. At each available target layer in L12-16, calculate centered Pearson correlation between $\hat{\kappa}_4$ and the teacher on the sensitive neurons $|\mu/\sigma|<1.5$, using the teacher mask. Require correlation $\geq$0.7 on both networks with no fitted rescaling or noise correction. Failure at any available target layer falsifies this birth model. Do not rescue it by fitting λ or depth coefficients.

Q → D: data coverage. The synthetic teacher has even layers plus L15, so it covers L12, L14, L15 and L16 here. A failure on these suffices to reject the model; passing them does not certify the missing synthetic L13. Passing the full stated gate requires that additional teacher slice.

D / E: test cost. If the baseline source buffers must be regenerated, two baseline passes plus this diagnostic cost approximately 464.7035 u [D from the approximate quoted baseline], excluding any missing-channel extraction. Runtime should be under one CPU hour on hardware that can run the quoted baseline [E; not timed here]. If the gate passes, the next validation is the ordinary fully coupled mean-closure run with $\hat{\kappa}_4$ replacing the regenerated diagonal; the correlation gate alone does not establish a final-MSE gain.

Scope: all benchmark measurements are quoted from the prompt. No teacher arrays were supplied to this session, so the network gate has not been executed.